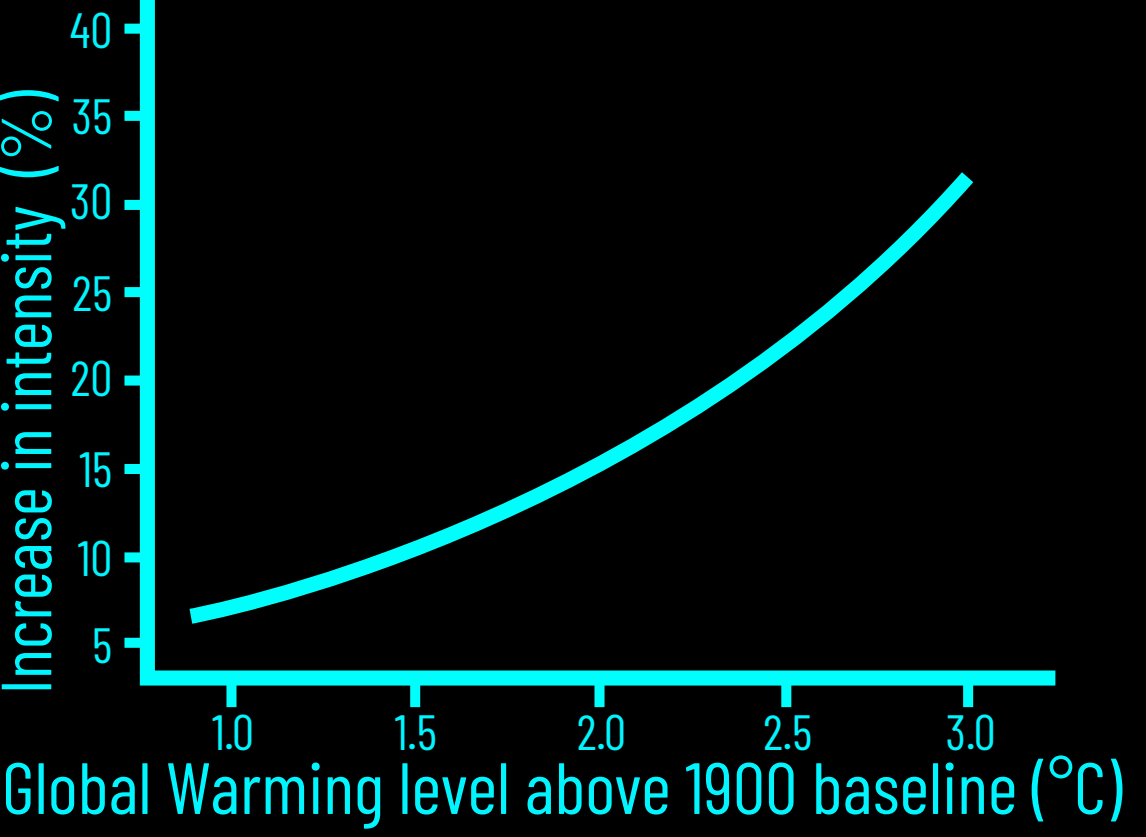


Anticipating flood induced morphological damages from high-resolution topography

Extremely high precipitation events



Typically, an extreme flooding event combines:

- 10s to 1000s of casualties and M\$ of damages
- 10s to 100s metres of erosion and aggradations over 100s of kms
- 100s to 10000s years of recovery time for the natural system

Recent example include Nepal (2021, 2026), La Berarde (France 2024), Alex (France, 2020), ...



Aftermath of the Melamchi flood (Nepal, 2021)

Some of the main controls on erosion and aggradation are the flood intensity and the **pre-existing topography** (e.g. river and valley width, slope, ...) controlling the spatial variation in shear stress (i.e. "river erosion potential").

Using flood simulations and high-resolution topographic data, can we predict the localisation and volume of erosion and aggradation for future events?

A methodological challenge

Catastrophic floods cascade erosion and aggradation events over 100s of kms with a contributing area of 1000s of km squares.

Topographic Data (Digital Elevation Models):
10e¹⁰ pixels to process
and 10s Gb per field (e.g. flow depth, velocity, ...)

Current flood models:
10s of hours of compute time for a single flood event over a few kms squares (10e⁶ pixels)

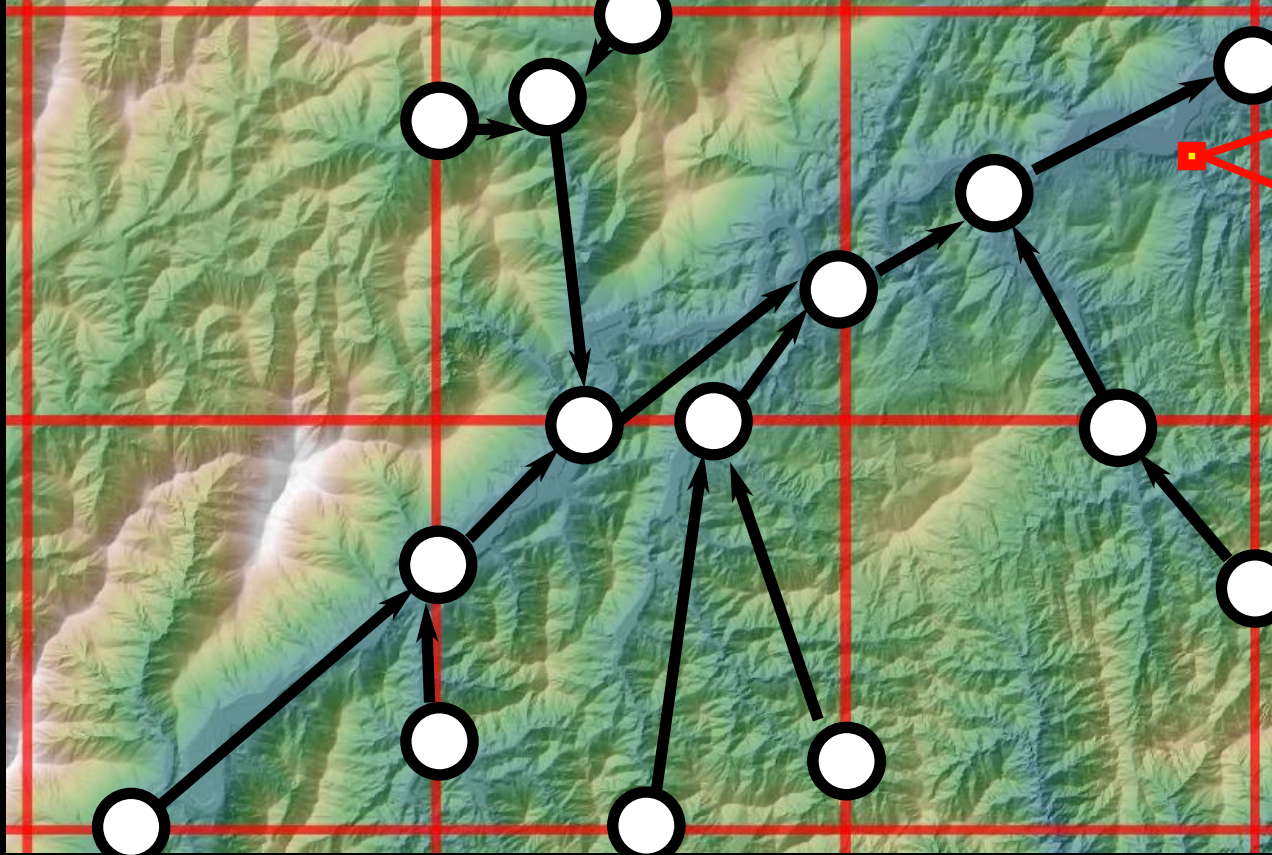
Novel approach

Sadly, the most impactful but the least graphic output of my research ...

2. Out of core flow accumulation:

Correcting local minima and approximating peak discharge without loading the whole DEM into memory

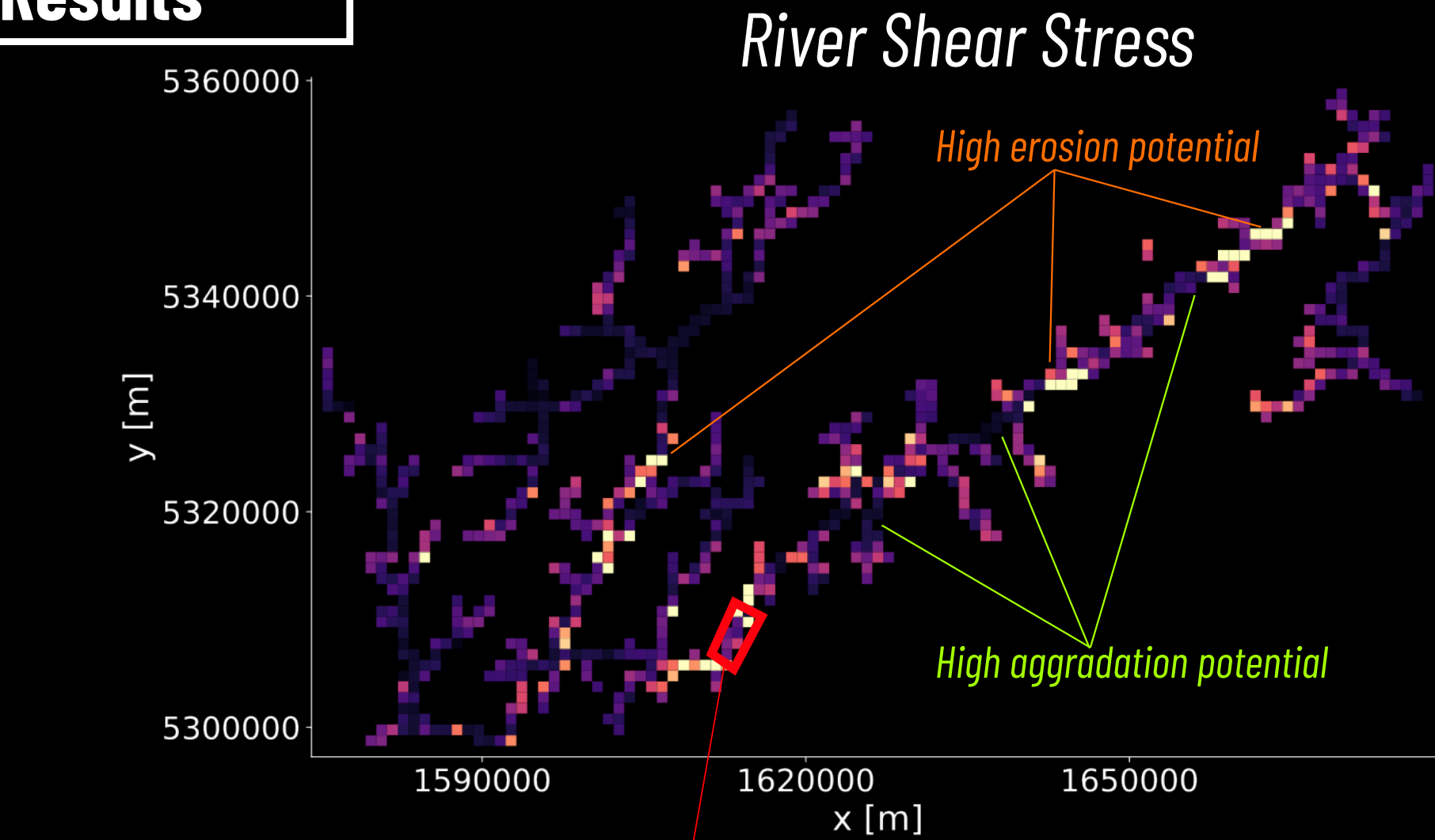
OOC = Out of Core. The DEM is split into tiles processed independantly. The challenge is that flow needs **global** information - e.g. flow line from the ridge to the outlet (the ocean) - so one need to "juggle" between **local** flow and **global** graph.



3. Multiscale flood modelling on GPUs

Using the global OOC approximation and adapting flood model to mass parallelism (x10 to 1000).

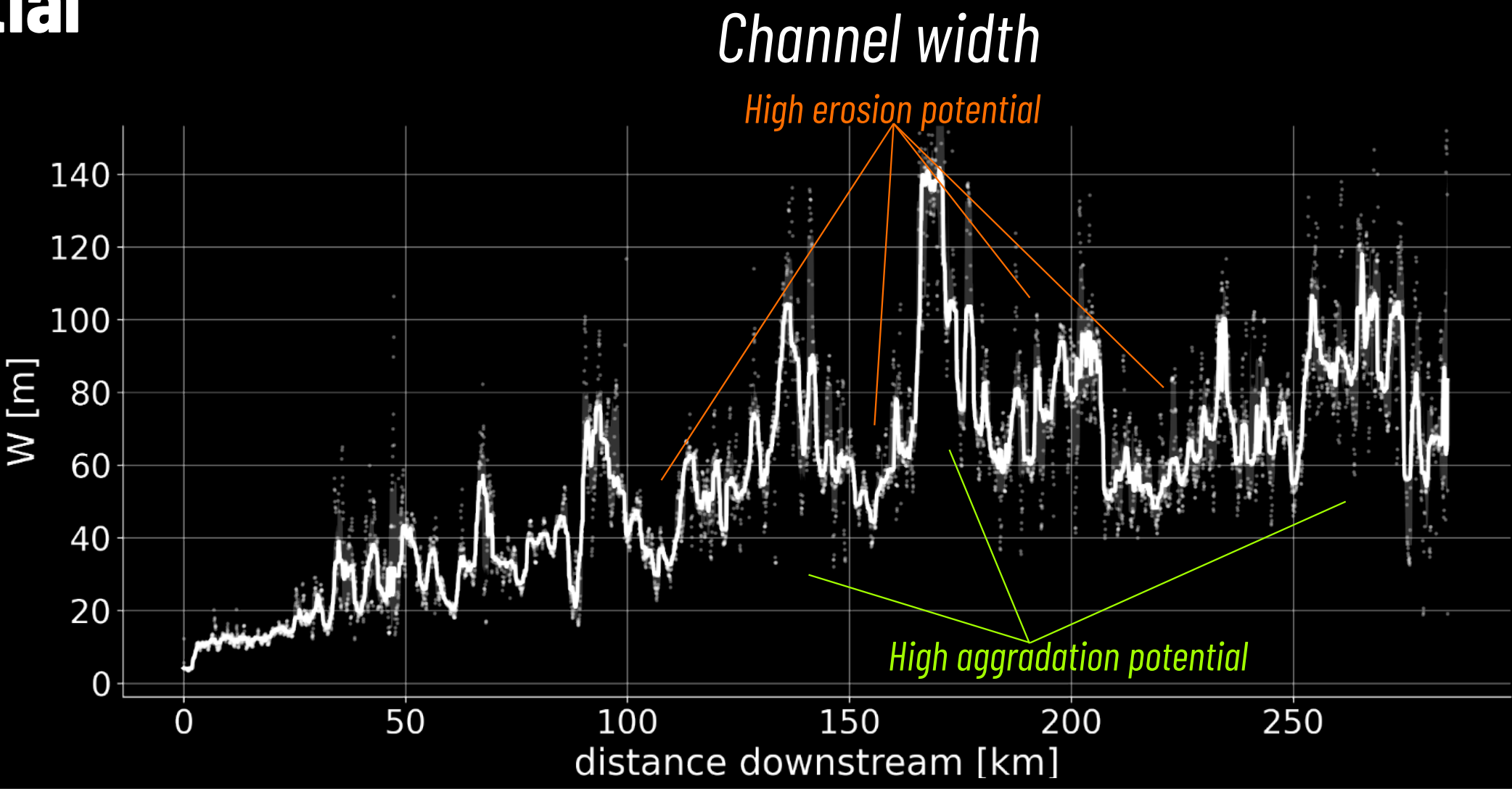
Results



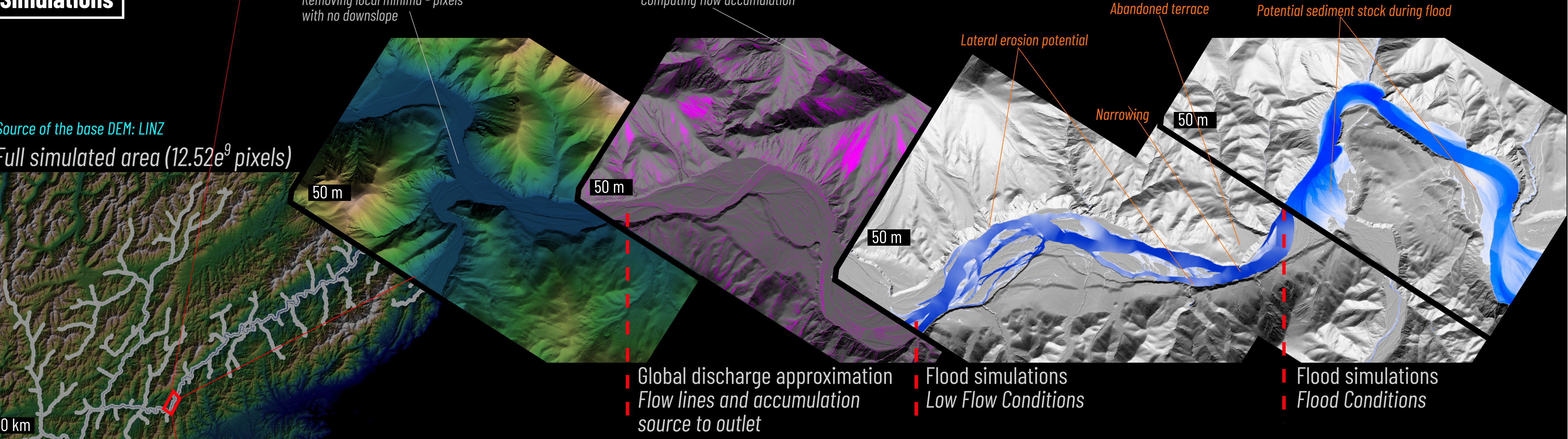
Erosion and aggradation potential

Combining width and shear stress data, computed from the physics of flood simulations, allow for suggesting potential "trap" for sediment (aggradation) and sources (zone where sediments gets eroded).

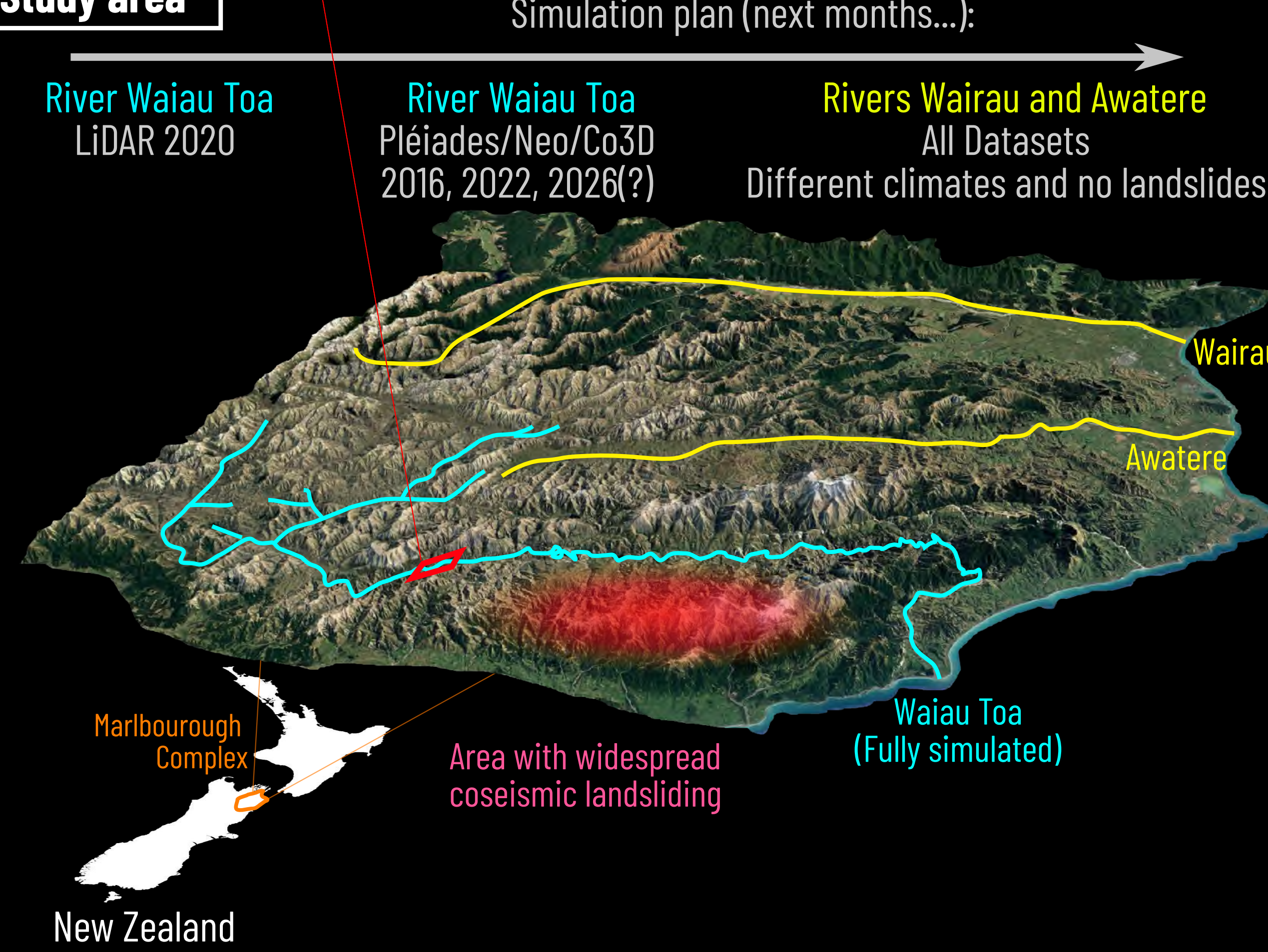
Next steps: integrating sediment flux to convert these potentials into volumes eroded - transported - aggraded



Simulations



Study area



The Marlborough region was strongly affected by the 2016 Kaikōura earthquake. The magnitude 7.8 event triggered more than **20,000 landslides**, supplying the Clarence River and its tributaries with large volumes of readily mobilizable sediment.

Pléiades, airborne LiDAR, Pléiades Neo, and CO3D provide more than a decade of high-resolution topographic observations, enabling the quantification of erosion and sediment transport across the affected catchments.

Combined with landslide inventories and climate data, these observations allow relationships between flood events, sediment sources, transport pathways, and sediment residence times to be investigated.

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Topotoolbox consortium
(topographic analysis with pytopotoolbox and GPU toolbox)

